BIO319 Coursework

Introduction

Imagine you are a data journalist working at a newspaper. You have been provided with two datasets with global health and economy data and are asked to put together an infographic. You have journalistic freedom on which story you would like to tell. You can combine these datasets with data found elsewhere.

Setting up your project

Just as in the workshops, accept the assignment invite on GitHub classroom, then create a project in PositCloud within the BIO319 workspace. The project contains these instructions and the datasets. You will need to add your script, your figure, and any external datasets you have used (extra datasets are optional).

The assignment

Your first task is to explore the data. Make sure to read the data dictionary below. What are the variables? Do both tables follow tidy rules? The data are patchy and not all statistics are available for all years and countries. You will have to tidy, subset, and join the tables for your data visualisation. **You will need to export your tidied data frames in a suitable format.**

You will then **produce a figure that contains at least two panels and tells a story**.

The first panel will show the relationship between economic status and life expectancy. You can decide whether to show selected countries or regions, one year or a time course, and which of the economic measurements to use (GDP, health spending, income share or income group).

For the other panel(s) you are free to choose. You will decide what story or observations you would like to share. This will need some preparation where you explore the data and come up with ideas. To get full points (see marking criteria below) you may also incorporate other datasets, for example from Our World in Data or Kaggle to support your story. Make sure to reference the source of these external datasets properly.

Some ideas for external data would be infectious disease rates, cardiovascular disease, happiness index, etc...

You will then use Kaltura software (instructions here) **to record an explanation of the plot and the code**. The maximum length is ten minutes (we will stop the video at the 10-minute mark and not grade beyond it). We want a spontaneous, not scripted, explanation to assess your understanding of your plot and code.

How to download Our World in Data datasets (optional)

It's very easy to download datasets from OWiD, see detailed instructions here. Put simply, search for your data of interest here and then replace anything after (and including) the question mark with ".csv" in the URL. For example:

https://ourworldindata.org/grapher/gdp-vs-happiness?time=2024..2024&tab=scatter

https://ourworldindata.org/grapher/gdp-vs-happiness.csv

To upload data to PositCloud click on "Upload" (green arrow in screenshot below).

Rules

- Your code should be clearly formatted and extensively commented while maintaining readability of the code itself.
- All plots should be *as far as possible* self-explanatory; titles and subtitles should be used to bring across the message.
- External datasets must be publicly available. You must provide a link to the dataset in your report. Your code needs to run as it is, including importing the external dataset.

We expect the following files from you:

- **One recording**

In a zip folder (see submission instructions below):

- **One R script** that contains the code to reproduce your cleaned data and graph. This R script should be runnable without any other modifications by the graders.
- **The original csv tables we provided**
- **Tidied data** in a suitable format (can be several tables)
- **One graph file** saved in a suitable format (e.g. .png)
- **One .txt file with your AI use declaration**
- **Optional**: any external datasets you have used (max file size 50 MB). Alternatively, your code may contain a line where you download the dataset.

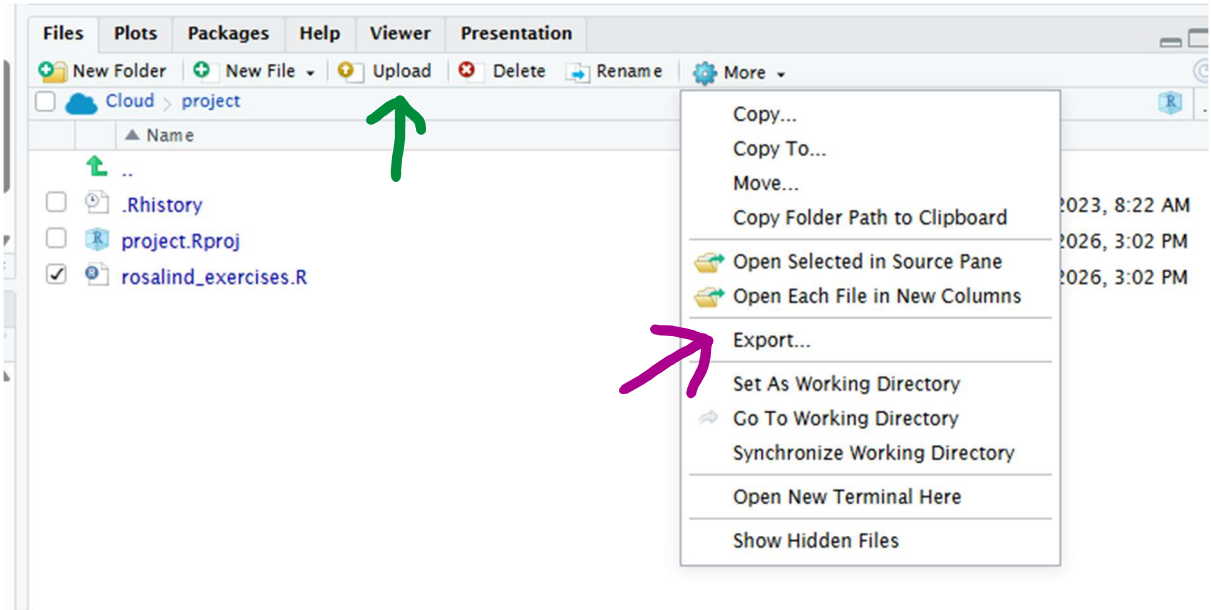
Marking criteria

1) **Recording**: Are cleaning and plotting explained well? Does it provide an interpretation and not only description of the figure? (15/100 marks)
2) **Code**: Does it run without errors and is it concise, readable and well commented? (15/100 marks)
3) **Tidied data**: Are the exported tables correct and tidy? (10/100)?
4) **Figure**: Are the panels conveying a clear message with appropriate design and geom choices? Do they look professional with bespoke colours and themes? (40/100 marks)
5) **External Data**: Does it use external data in a meaningful way and are the data provided? (20/100 marks)

Submission

The assignment is released Friday 27th March via email and due Monday 13th April 17:00. You will submit your assignment via QMPlus.

To export your files from PositCloud click the box next to the files you'd like to export and then export in the "More" menu. See purple arrow in screenshot below. This produces a zipped folder. **Submit the zip file and your recording on the assessment tab.**



Data dictionary

demographics.csv

Variable	Description
Entity	Countries and “World”
Code	Country Code
Year	Year
Population	Population
GDP.per.capita	Yearly gross domestic product
World.region.according.to.OWID	World region
CHE.as.percentage.of.GDP	Current health expenditure as % of GDP
Richest.1.	Income share (in % before tax) held by richest 1%
Next.9.	Income share (in % before tax) held by next 9%
Middle.40.	Income share (in % before tax) held by middle 40%
Poorest.50.	Income share (in % before tax) held by poorest 50%
Income_group	World Bank income classification 2026

health_data.csv

Variable	Description
Entity	Countries and “World”
Code	Country code
Year	Year
Life.expectancy	Life expectancy, both sexes
Alcohol.consumption	Pure alcohol consumption in litres/year, per person >15 yrs , both sexes
Concentrations.of.fine.particulate.matter..PM2.5	Fine particulate (PM2.5) air pollution in $\mu g/m^3$
Prevalence.of.obesity.among.adults..BMI....30..	Obesity (BMI>30) in % in adults >18 yrs, both sexes
Hepatitis.B..HepB3.	Share (%) of one-year-olds who have been immunised against hepatitis B.
H..influenza.type.b..Hib3.	As above
Inactivated.polio.vaccine..IPV1.	As above
Measles..first.dose..MCV1.	As above
Pneumococcal.vaccine..PCV3.	As above
Polio..Pol3.	As above
Rubella..RCV1.	As above
Rotavirus..RotaC.	As above
Diptheria.tetanus.pertussis..DTP3.	As above